

Stats 4A03: Time Series
Final Project Report

Forecasting Groundwater Depth Using Time Series Modeling Techniques

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1 Introduction

Water is one of the essential natural resources for human activities. It not only satisfies our demands for drinking, but also plays a critical role in agricultural irrigation, industrial production, and ecosystems. People have realized the importance of water resources and are willing to take action to protect them. However, compared to visible rivers and lakes, groundwater, which lies under the surface of the earth, has not received the attention it deserves. Groundwater is commonly used for agricultural irrigation and in some areas it is the water resource that people use for daily living [1]. Therefore, groundwater is an important part of daily life, and changes in its depth can directly affect agricultural productivity and water availability. Since groundwater is invisible, it is difficult to confirm the exact depth. However, we can use tools to predict the depth of groundwater.

The project aims to use time series modeling to predict the depth of groundwater based on historical data. The prediction of groundwater resources can help us to better manage the use of water resources and maintain a sustainable water ecosystem.

2 Modeling

2.1 Data Description

The data include hydrological and climatological measurements located in Petrignano, Italy [2]. For this project, the focus was on the relationship between time and groundwater depth.

Before the modeling started, we filled in the missing values in the data to ensure continuity. The depth to groundwater was not recorded at some points in time, so we used the Last Observation Carried Forward (LOCF) to fill in the missing information and ensure the completeness of the data [3].

In addition, the dataset records groundwater level information at two different monitoring locations. In order to provide a more comprehensive representation of groundwater depth at a given time, we obtained an average for both locations at the same time as the representative groundwater depth at that time. Besides, in order to simplify the model structure and reduce the influence of data fluctuation, we chose to do the modeling in a time unit of month. In other words, we calculated the average depth of groundwater for all observations in each month and used it as the depth of groundwater for that month to input into the time series model.

After feature engineering and removing unused columns, we get a dataset with 172

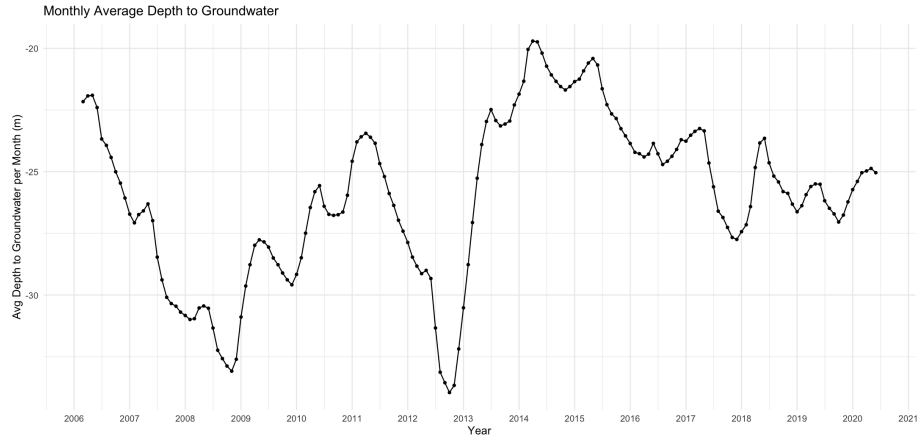


Figure 1: Monthly Average Depth to Groundwater

observations and 2 variables. We will build our modeling based on these data.

2.2 Tentative Model

According to the definition of the autoregressive integrated moving average (ARIMA) and multiplicative seasonal autoregressive integrated moving average (SARIMA) models, we need to build a proper model based on stationary data. First, we use the Augmented Dickey-Fuller (ADF) test to check whether the original data is stationary or not. The most common value used for significant level is 0.05 and we will also choose this value as significant level. Based on Table 1, by ADF test, the p-value for the original data is 0.2196, which is greater than the 0.05 significance level. This indicates that the original series is non-stationary. After applying first-order and seasonal differencing, the p-value decrease to 0.01133. This indicates that the transformed series is stationary.

Table 1: Augmented Dickey-Fuller (ADF) Test

Data	p-value
Original Data	0.2196
Transformed Data (first and seasonal difference)	0.01133

Therefore, in order to obtain stationary data, we need to use some methods to remove the non-stationary part. At first, we difference the data to the first order, but the difference series still showed a seasonal pattern, which indicates that there is still seasonal non-stationary in the data. To remove this, we apply another seasonal difference with

period 12 to the data that have already been first-order differenced. After completing this, we run the ADF test again and find that the differenced data are stationary since the value of p-value is less than 0.05. This implies that after one first-order differencing and one seasonal differencing, we have stationary data that can be used for time series modeling.

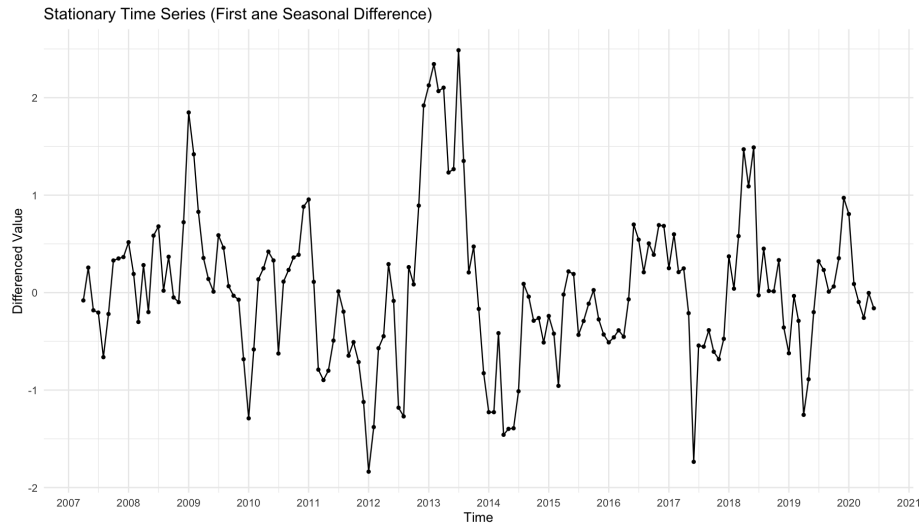


Figure 2: Stationary Time Series

Next, we need to determine the order of the time series model based on autocorrelation function (ACF) and partial autocorrelation (PACF) plots of the transformed data. Based on Figure 3, we observe that in the ACF plot there is no spike present indicating a zeroing trend after some particular lags. This indicates the absence of any moving average (MA) component. Instead, the ACF gradually decays to zero, which implies the existence of the autoregressive (AR) component. In addition, for PACF, we observe that it zeros out after lag 1, which indicates the existence of AR(1) component in the model. Although there exist some spikes in some following lags, they do not suggest a clear seasonal pattern. For now, we just ignore it and consider it as the result of noisy data.

In conclusion, we tentatively model these data as $SARIMA(1, 1, 0) \times (0, 1, 0)_{12}$ based on the definition.

2.3 Model Diagnostic

After setting up the model, we would like to assess the quality of our fitted time series model by different methods. One of the most common methods is to check the properties

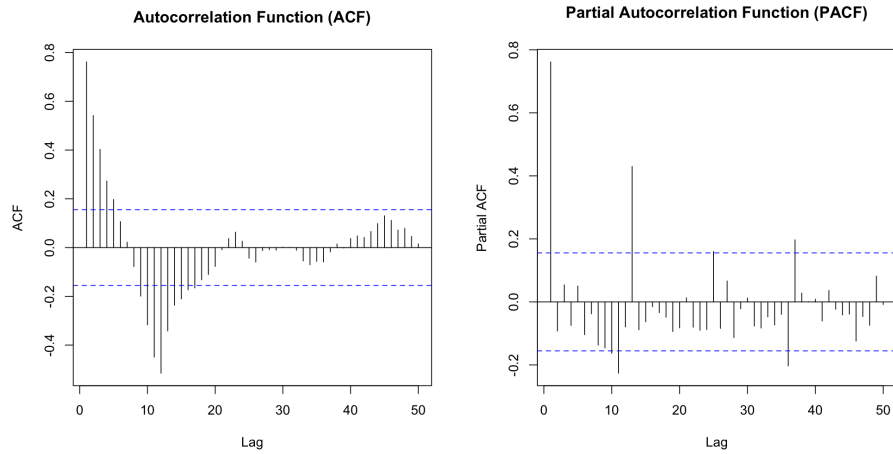


Figure 3: ACF and PACF for Stationary Time Series

of the residual. If our model is good fit, then the residuals should behave approximately as white noise.

Based on Figure 4, we can observe that the standardized residuals fluctuate around the zero mean without any observable trend. In other words, the model is relatively well behaved, which implies that the model has captured the structure in the data. Also, in the ACF plot of the residuals, we notice that most of the ACFs lie within the dashed line, which by default represents the 0.05 level of significance. So, it shows that most of the ACF estimates for the most lags do not exceed the 95% confidence interval. This indicates that the residuals are mostly uncorrelated across lags, which is consistent with the properties of white noise. Additionally, the last plot in Figure `reffig::tsdiag` focuses on the independence of the residuals in the joint lags. According to the results of the Ljung-Box test, most of the p-values are above the dashed line, which implies that they are above the significance level. This suggests that for most of the joint lags, they cannot reject the original hypothesis that the residuals are white noise, so there is no significant autocorrelation.

Furthermore, for normality, we could check that by plotting a histogram of the standardized residuals. According to Table 5, we observe that the histogram is approximately bell-shaped. This is consistent with what we expected if the data are normal distributed. In addition, we could also check it more carefully by plotting the quantile-quantile (QQ) plot. The QQ plot should look approximate like a straight line if the data is normally distributed. Although we observe some outliers at the ends in Figure 6, the overall distribution of points remains close to a straight line. We conclude that the

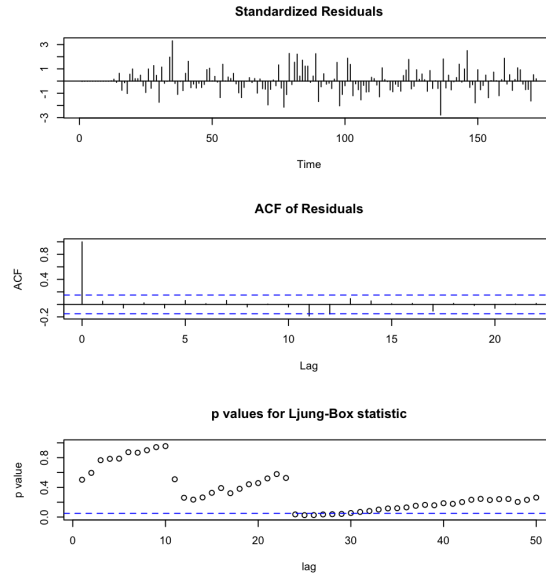


Figure 4: Residual Diagnostics

residuals are approximately normal distributed.

In addition, we overfit our data with a slightly more general model. In this case, we choose $SARIMA(2, 1, 0) \times (0, 1, 0)_{12}$ as our overfit model for our data. We determine whether our tentative model is adequate or not based on the estimates of parameters for the overfit model. Compared with the parameter estimation results of the two models, it can be found that the shared parameter remains consistent with the addition of the ϕ_2 . According to Table 2, we find that the estimate of ϕ_1 changes from 0.7579 to 0.8289. Since there is no significant difference between these two values, this implies that the estimation of this parameter is consistent. On the other hand, the estimate of the added ϕ_2 is -0.0925, which is close to zero, which suggests that this parameter is not statistically significant. This result shows that the added ϕ_2 in the overfit model has a limited contribution to the model improvement, and that the original model without the non-zero ϕ_2 term, the tentative model, is adequate.

Table 2: Overfitting

Model	ϕ_1	ϕ_2
Tentative Model $SARIMA(1, 1, 0) \times (0, 1, 0)_{12}$	0.7579	
Overfit Model $SARIMA(2, 1, 0) \times (0, 1, 0)_{12}$	0.8289	-0.0925

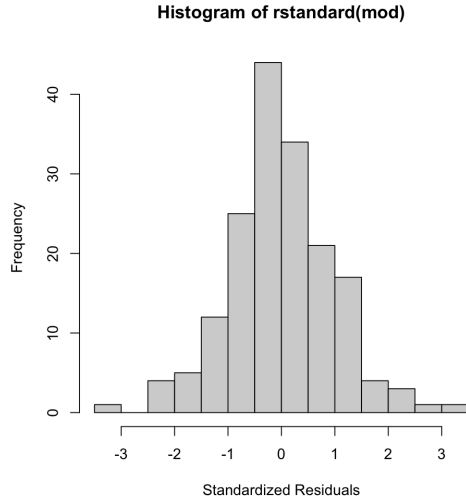


Figure 5: Histogram of Standardized Residuals

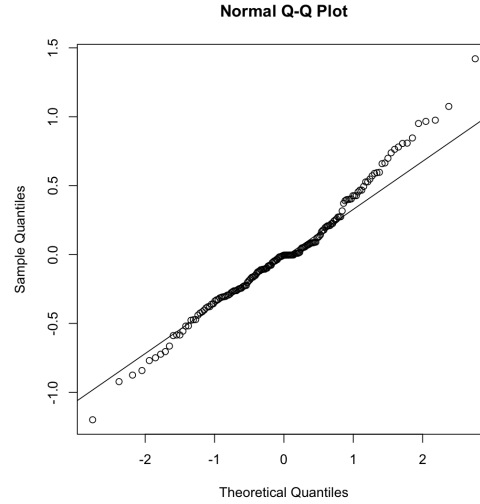


Figure 6: QQ Plot of Residuals

In conclusion, this seasonal tentative model is a reasonable good fit.

3 Results

Based on the adequate model, we can predict the future depth change of groundwater, and the results we obtained are shown in Figure 7. It is important to note that the standard error of the prediction increases exponentially with time according to the Figure 7, which indicates a significant increase in the uncertainty of the long-term prediction. In other words, in the short-term prediction, the model shows high reliability, while in the long-term prediction, the uncertainty increases significantly.

The prediction of the depth to groundwater is very useful for the appropriate management of water resources. For example, since groundwater is widely used for agricultural irrigation, we can adjust our irrigation schedules based on predictions of the depth of groundwater in the coming months[1]. When the prediction suggests that the depth of groundwater will decrease beyond a certain value in the coming months, it means that we will have a hard time getting enough groundwater for irrigation. In this case, we can start a water-saving irrigation scheme earlier. This avoids over-consumption of water and ensures stable agricultural production.

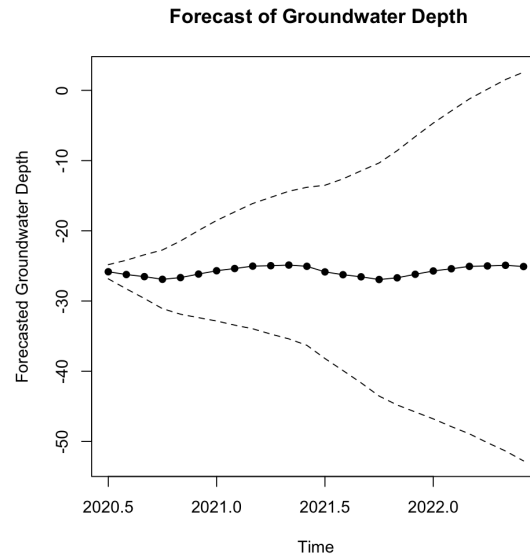


Figure 7: Forecasting

In addition, for people in some areas, groundwater is the only source of water for their daily lives[1]. The change of groundwater depth is closely related to their daily life. The decrease of groundwater depth to a certain level means that it is difficult to get enough water. Identifying the risk of a decline in groundwater depth months into the future through forecasting models allows communities to prepare emergency plans, such as stocking up on drinking water and using alternative sources of water that can keep people's lives unaffected.

Therefore, the prediction of the depth to groundwater can help us to better respond to changes in groundwater.

4 Conclusion

This model is useful in predicting changes in groundwater depth, but it still has some limitations. First, the model is not reliable for long-term predictions, but it is useful for short-term predictions. It is recommended to use a dynamic forecasting method, such as rolling forecasts, for practical applications[4]. This means that the forecasts are not made at a fixed point in time each year, but are rolled forward on a monthly basis, forecasting up to 24 months into the future[4]. By rolling monthly forecasts, long-term forecasts can be

broken down into successive short-term forecasting processes, so that the actual application is always based on highly reliable forecasts of the last few months, which effectively avoids the risk of uncertainty in long-term forecasts.

Also, the model does not consider the effects of other factors on groundwater depth, such as weather extremes. The drought may lead to a rapid decrease in groundwater depth, while heavy rainfall may lead to a rapid increase in it. This limitation may lead to significant deviations in the prediction results in practical applications. Future improvements will need to combine weather data to find a connection between weather extremes and groundwater depth to improve the precision of the model.

Finally, the model does not account for differences due to geologic structures. The ignoring of these geological factors may significantly reduce the precision of the model when it is applied to other regions. Future improvements will need to combine geological data to find the connection between geologic structure and groundwater depth to improve the regional applicability and precision of the model.

In conclusion, the construction of this time series model for forecasting groundwater depth has provided us with a deeper understanding of the application of time series analysis in the simulation of hydrological systems. Although the model is able to reflect the general trend of groundwater levels relatively well, there are still several important limitations that need to be improved.

References

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